



General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	04/07/2025
Assigned Reviewer	Samuel Prieto Ayllon
Next Review Date	04/07/2026

Signed Off By	Oraib AlKheetanRojas
Signed Off On	19/05/2026
Sign Off Comments	

Operation Assessed	Robot End-Effector – Robotiq 3-Finger Adaptive Gripper Operation
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Associated with specific area

 Basement 2

 B2 029 (Robotics Lab)

 CTP

Description of work area and/or activity assessed

The activity covers routine installation, programming and use of the Robotiq 3-Finger Adaptive Robot Gripper mounted on an industrial or collaborative robotic arm in the lab. Work involves gripping, manipulating and releasing test objects of varying size and mass under supervised research conditions. Power is supplied at 24 V DC (SELV) via a dedicated bench top supply; communication is through Ethernet/field-bus cables routed along the robot harness. Operations take place indoors on a level, non-slip floor.

Reason for Risk Assessment

To document and control health, safety and equipment-damage risks introduced by using the 3-Finger Adaptive Gripper in active research.

Overall Current Risk

Moderate Risk



Average Number of Persons Affected

2  Employees

2  Students

2  Visitors

Current Rating	Hazard Information	Task Status
Moderate Risk 	Pinch Points The three articulated fingers and scissor axis move rapidly during closing and mode changes. Fingers can exert up to 60 N and automatically adapt around objects, creating multiple pinch or crushing points. Accidental contact could injure hands, entangle clothing or trap loose items.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Mandatory hardware & software emergency-stop integrated with robot. - Gripper motion limited by robot safety settings (speed & force). - Operators trained to keep hands and clothing clear of the gripper while powered.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	Electricity The gripper is powered by 24 V DC = 2 A. Although classed as Safety Extra-Low Voltage, damaged insulation, incorrect wiring or substitution with an AC source could cause mild shock, short circuit or sparks during installation.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Certified 24 V DC power supply with 4 A fuse; “power-on-last” rule enforced. - Cables strain-relieved and routinely inspected for wear. - Prohibition on supplying the gripper with any AC source. - Only authorised personnel may service electrical connections.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	Projectiles If grip force or geometry is inadequate an object may be dropped or even catapulted outward during fast robot moves, posing impact hazards to personnel or equipment.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Grip parameters (mode, position, speed, force) verified in teach mode before automatic runs. - Work piece mass < payload limits; secure trajectories below shoulder height where practicable. - Eye protection required for observers. - Collision-detection and force-torque monitoring stop the robot on abnormal load spikes.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.